

Digital Signature

Public key

How can we achieve both confidentiality and authentication?

RSA Digital Signature

↓
For digital signature scheme set up is
done by sender (ALICE)

$$Y = E_d(m) = m^d \bmod n$$

ALICE
m

BOB

$$e_A = (n, e)$$

setup

RSA Decryption using e_A

$$= Y^{e \bmod n}$$

$$= [m^d]^e \bmod n$$

$$= m^{ed} \bmod n$$

$$= m^{k\phi(n)+1} \bmod n$$

$$= \underbrace{m \bmod n}_{m} \underbrace{[m^{k\phi(n)+1}]}_{1}$$

$$= \textcircled{m \bmod n}$$

$$= m$$

1. p, q

2. $n = p \cdot q$

3. $\phi(n) = (p-1)(q-1)$

4. e such that
 $\text{GCD}(e, \phi(n)) = 1$

5. find d such that
 $e \cdot d \equiv 1 \bmod \phi(n)$

$$K = (n, p, q, e, d)$$

$$e_A = (n, e)$$

$$d_A = (p, q, d)$$

use for signature

Encryption scheme

$$(P, C, K, E, D)$$

$K \in K \quad \exists e_k \in E$
and $d_k \in D$ s.t. that

$$D_k(E_k(m)) = m$$

E_k and D_k should be computationally feasible function

Signature scheme

(P, A, K, S, V)
plaintext space

signature space
(set of all possible signatures)

keyspace
set of all possible signature algo.
verification algo.

such that -

For each $K \in K$
 $\exists \text{sig}_K \in S$
and $\text{ver}_K \in V$

$$\begin{aligned} \text{Sig}_K &: P \rightarrow A \\ \text{Ver}_K &: P \times A \rightarrow \{T, F\} \end{aligned}$$

$$\text{Ver}_K(x, y) = \begin{cases} T & \text{if } y = \text{Sig}_K(x) \\ F & \text{otherwise} \end{cases}$$

for every signature we should be able to verify

Digital Signature

K_S and K_V
in place of K_{priv}
and K_{pub}

Cryptography

Symmetric

- DES
- AES
- Stream

How to use?

Asymmetric

- RSA
- DLP
- ELGAMAL
- ECC

Protocols

- Digital signatures
- Security services
- RSA digital sign
- Attack against RSA d.s.

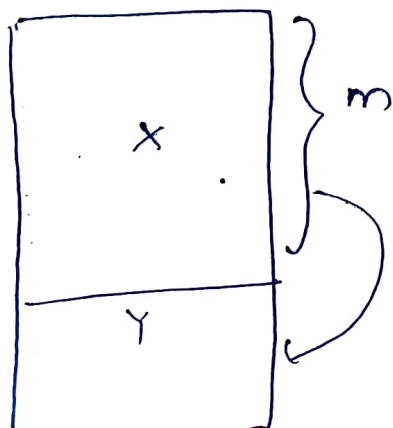
① Introduction to Digital Signatures

Goal - Signature function for the electronic world

conventional (paper) signature

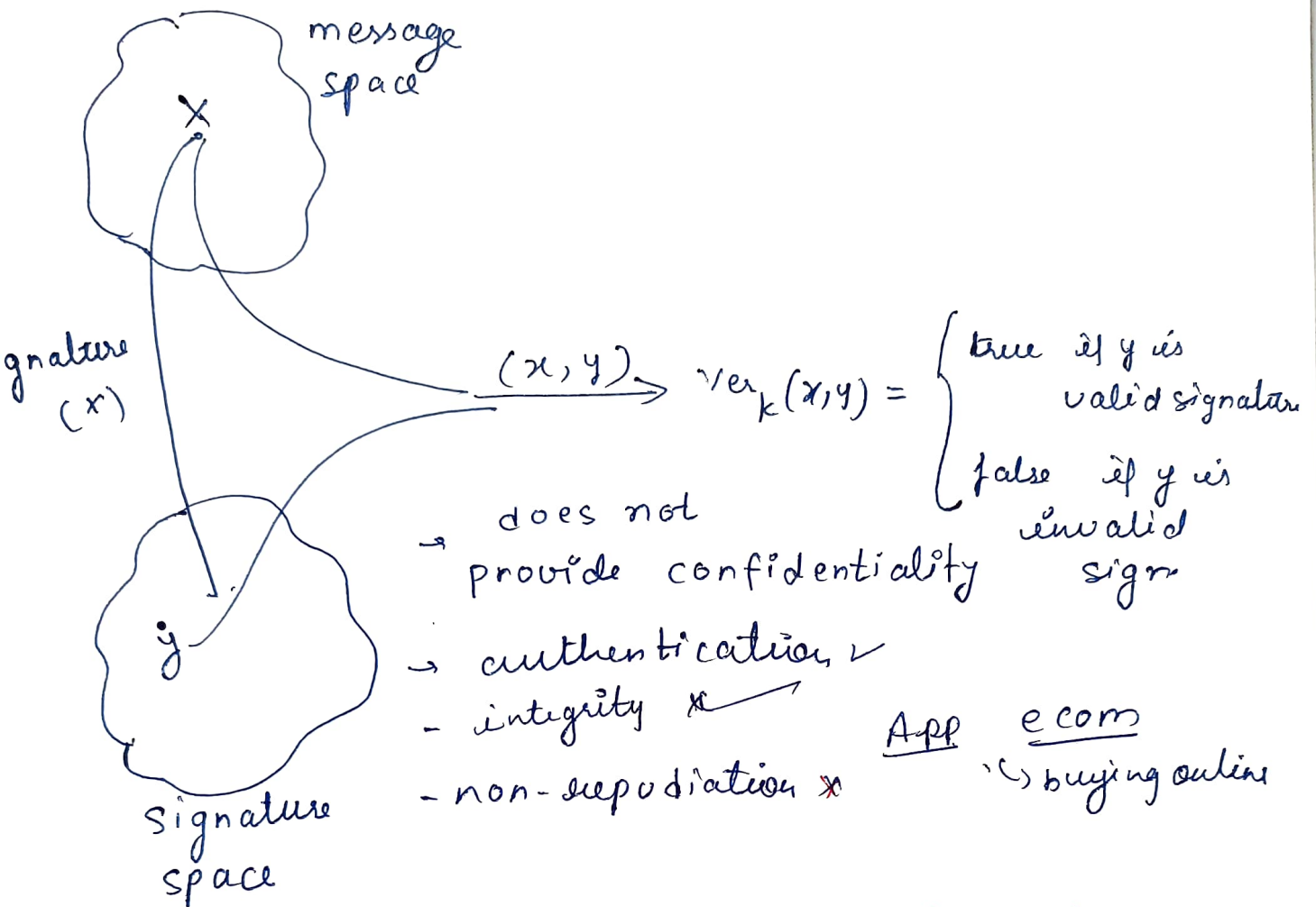
(Authentication)

let's try same digitally → 1000 bits



$$\text{Sig}_{K_S}(x) = Y$$

Signature



Security Services → The objectives of a security system are called "Security services"

Confidentiality

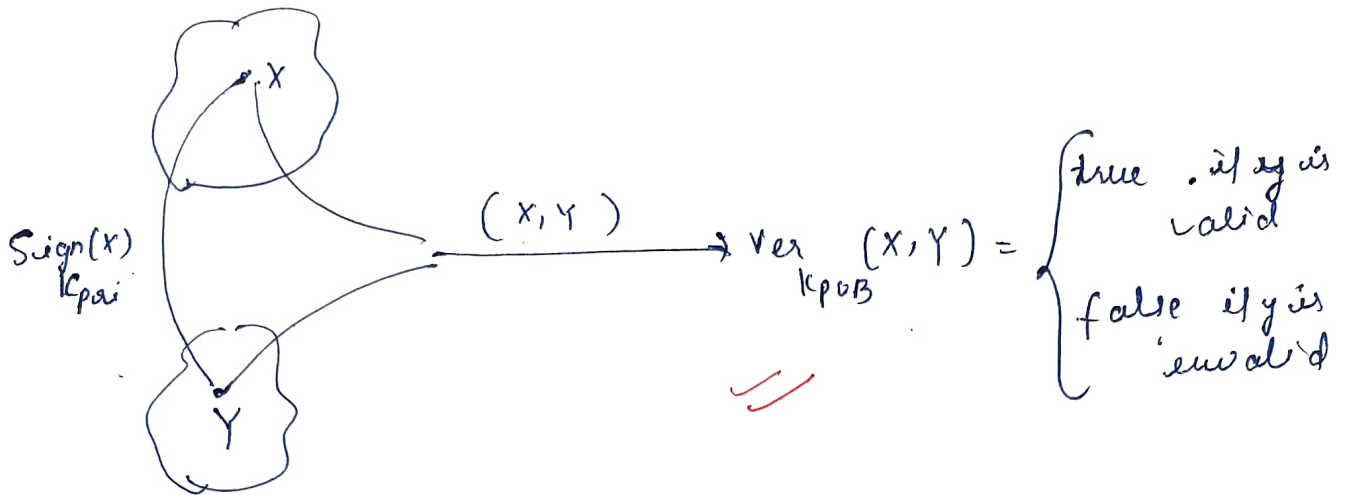


Message Authentication: sender of message is authentic ✓

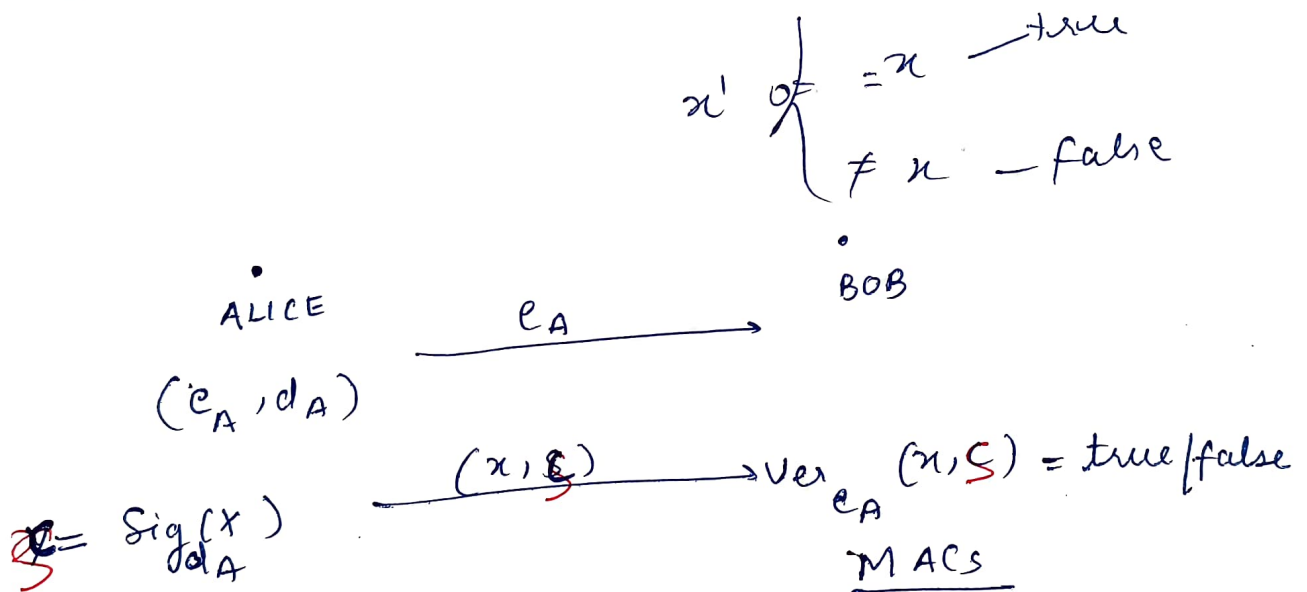
Integrity - message has not been modified during transmission ✓

Non-repudiation The sender of message cannot deny the creation of msg.
(A or B are cheating)
sender or receiver

Asymmetric Cryptography



Basic protocol with digital signature



RSA digital signature

set up

$$e_A = (e)$$

$$d_A = (d)$$

$$e_A = (n, e)$$

Protocol

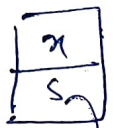
$$\text{Alice} \xrightarrow{(n, e)} \text{Bob}$$

$$e_A = (n, e)$$

$$d_A = d$$

$$S = \text{signature}$$

$$S = \text{Sig}_{d_A}(X) \equiv X^d \pmod{n}$$



$$\begin{aligned} &\text{Sig}_{d_A}(S) \\ &\text{Ver}_{e_A}(X, S) \\ &S \pmod{n} \\ &= X' \pmod{n} \end{aligned}$$

RSA signature scheme

p, q two prime

$$P = A = \mathbb{Z}_n \quad n = pq$$

set of all possible signatures

$$K = \{ (n, p, q, e, d) \mid ed \equiv 1 \pmod{\phi(n)} \}$$

$$\gcd(e, \phi(n)) = 1$$

$$K_{\text{pub}} = \{ n, e \}$$

$$K_{\text{pri}} = \{ p, q, d \}$$

$$x \in \mathbb{Z}_n \quad \text{sig}_K(x) = x^d \pmod{n}$$

Verification]

$$\text{Ver}_K(x, y) = \text{true} \quad \text{iff} \quad x \equiv y^e \pmod{n}$$

$$(x, y) \in \mathbb{Z}_n$$

Proof of correctness

$$s^e = x^{ed} \equiv x \pmod{n}$$

computational aspects

① Signing - square and multiply same algorithm (costly) - 18c

② Verification $s^e \pmod{n}$ in practice often

$$e = 3, \boxed{2^{16} + 1}$$

popular 17 bits

verification is fast

An attack against RSA

1. existential forgery attack against RSA

protocol $\downarrow$ OSCAR

ALICE

(n, e)

(x, s)

$$x \equiv s^{ed} \pmod{n}$$

$$x \equiv s^{ed} \pmod{n}$$

(x, s)

BOB

$$s^e \pmod{n}$$

$\downarrow$ = good

x

Verify

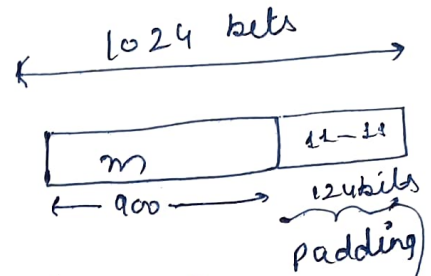
$$s^e \pmod{n}$$

$$\equiv x'$$

if $x' = x$ true

$x \neq x'$ false

In practice impose formatting rules on x can be checked by ALICE



$$Oscar: x \equiv s^e \pmod{n}$$

has a chance

2^{-124} to have a msg x 124 trailing 1's

$$2^{124}$$

tries

1. choose $s \in \mathbb{Z}_n$
2. compute $x \equiv s^e \pmod{n}$

BOB

$$e_B = (n, e), e_B = d$$

$$s \equiv x^d \pmod{n}$$

limitations of Hack - Oscar cannot directly control the semantic of the msg x .

Q Does the verification check out?

$$x' \equiv s^e \pmod{n} \text{ (Alice)}$$

$$x \equiv s^e \pmod{n} \text{ (Oscar)}$$

valid sig.

valid sig

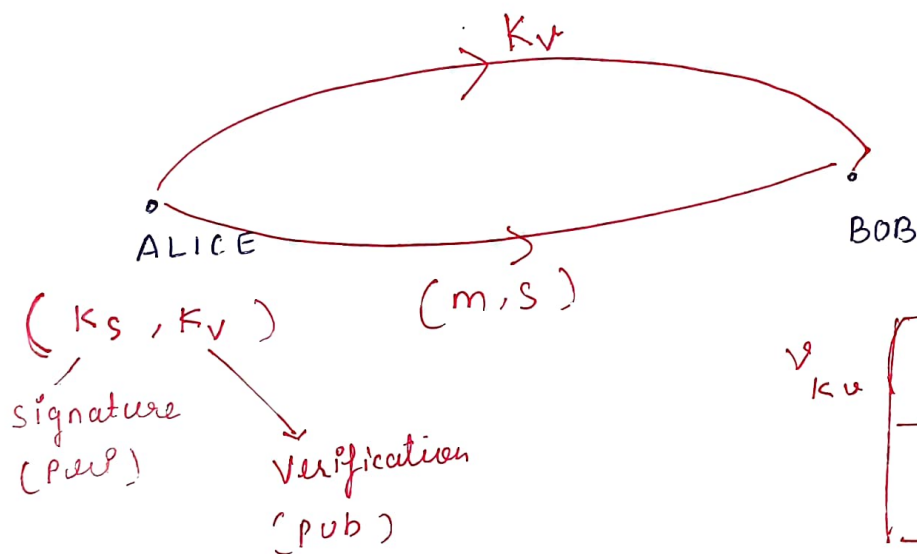
RSA Signature

A signature system is obtained from RSA by defining

$K_S = (n, b)$ and $K_V = (n, a)$
key for signature verification key

$$S_{K_S}(m) = (m, m^b \bmod n) \quad \text{and} \quad V_{K_V}(m, s)$$

$$= \begin{cases} \text{TRUE} & \text{if } m \equiv s^a \bmod n \\ \text{FALSE} & \text{otherwise.} \end{cases}$$



message m

$$S_{K_S}(m) = s$$

Faking this signature is one way or another is equivalent to breaking RSA.

An outside party can calculate some $s = m^a \bmod n$ (as the msg)

Blind signature If A wishes to sign a message m of B without knowing its content,

RSA Signature

$$K_s^A = (n, b) \\ \text{private}$$

$$\text{and } K_v^B = (n, a) \\ \text{public}$$

$$S_{K_s}(w) = \left(w, \frac{w^b \bmod n}{u} \right)$$

$$V_{K_v}(w, u) = \begin{cases} \checkmark & w = \frac{u^a \bmod n}{x} \\ \times & \end{cases}$$

Faking Signature (breaking RSA)

let say OSCAR select u such that

$$u = \frac{w}{x} \bmod n$$

$$w = u^a \bmod n$$

message.

This message does not contain any info.

$$S_{K_s}(w) = (w, \frac{w^b \bmod n}{x})$$

$$V_{K_v}(w, u) = \begin{cases} \checkmark & \frac{w^b \bmod n}{x} = u^a \bmod n \\ \times & \text{otherwise} \end{cases}$$

RSA can also be used to get a so-called blind signature. If A wishes to sign a message w of B, without knowing its content, the procedure is following:

1. B chooses a random number d such that $\gcd(d, n) = 1$, computes the number $t = (d^a w \bmod n)$ and sends it to A
2. A computes the signature $u' = (t^b \bmod n)$ as if message would be t and sends it to B.
3. B computes $u = d^{-1} u' \bmod n$.

$$d^{-1} u' = d^{-1} t^b = d^{-1} d^{ab} w^b \equiv \underline{d^{-1} d} w^b \equiv \underline{w^b \bmod n}$$